Exact Newform-Period Taxonomy for a Level-11 Time Change of the Modular Geodesic Flow

Liang Wang
School of Artificial Intelligence and Automation, Huazhong University of Science and Technology
Luoyu Road 1037, 430070, Hubei, P.R. China
wangliang.f@gmail.com

28 August 2026

Abstract

We study a positive time change of the geodesic flow on the level-11 modular surface whose first period variation is the real period of the unique normalized weight-two newform. The motivating question is whether the Hecke action on these periods can induce a primitive-orbit Euler recurrence. We first identify the correct owner of the Hecke relation: the pushforward of an oriented closed cycle, which is generally a sum of closed geodesic owners rather than a single primitive orbit. In the canonical inverse-closed orbit product, odd variations cancel. At second order, an all-parameter scalar recurrence is equivalent to exact degree-wise quadratic moment conditions. We then give a computer-assisted but exact classification of the frozen finite output multiset. A rational Schreier model of $H_1(Y_0(11),\mathbb{Q})$ has coordinates (x,y,z), real involution $\tau(x,y,z)=(-x,y+z,-z)$, and real-period coordinate $k=2y+z$. This makes every normalized period ratio rational and every quadratic moment a rational sum of squares. Among 138 Hecke cycle-owner instances, exactly two are full complex-period kernels, two are real-projection-only kernels, and 134 are true nonkernels. The candidate laws $\lambda_p=a_p$ and $\lambda_p=a_p^2$ each fail exactly 51 of 55 frozen groups; the control a_p^2-p fails all 55. These results refute the tested naive Hecke-like recurrences on the registered finite ledger, but they do not establish a global determinant, a prime–orbit correspondence, or a Route-A dynamical-zeta match.

Keywords: modular geodesic flow; newform periods; Hecke correspondence; time change; Schreier homology; dynamical zeta obstruction

繁體中文摘要

本文研究模曲面 $Y_0(11)$ 測地流的一族正時間變換，其閉軌道週期的一階變分由唯一正規化權二新形式之實週期給出。核心問題是：Hecke 對應在線性週期上的特徵關係，能否提升為逐一原始軌道的 Euler 型遞推。研究首先證明，正確的 Hecke 所有者是有向閉圈的推前，通常分解為多個閉測地線，而非單一原始軌道。標準的逆向封閉乘積使奇數階變分成對消失；二階恆等式則等價於按 Hecke 圈長分組的精確平方矩條件。本文以有理 Schreier 同調模型完成有限分類：$H_1(Y_0(11),\mathbb{Q})$ 中的實結構為 $\tau(x,y,z)=(-x,y+z,-z)$，實週期只取決於 $k=2y+z$。因此所有正規化週期比與平方矩皆為有理數，可完全排除浮點零判定。凍結資料中的一百三十八個 Hecke 圈所有者，恰有兩個完整複週期核、兩個僅實投影核及一百三十四個真非核；候選律 a_p 與 a_p^2 各在五十五組中精確失敗五十一組，控制律 a_p^2-p 則全數失敗。此結果是有限輸出多重集合上的精確否定定理，不代表完整原始軌道普查，也不建立質數對應、全域動力行列式或 Hilbert–Pólya 實現。

關鍵詞： 模測地流；新形式週期；Hecke 對應；時間變換；Schreier 同調；動力 zeta 障礙

1 Introduction

Arithmetic surfaces offer a natural place to ask how algebraic correspondences interact with periodic-orbit expansions. On the level-11 modular curve, the unique normalized weight-two newform supplies a canonical holomorphic differential, and its real part can be used as the infinitesimal clock change of the geodesic flow. The resulting period variation is intrinsic: it is an integral over an oriented closed geodesic, is invariant under subgroup conjugacy, changes sign under orientation reversal, and scales linearly under traversal repetition. These properties make the construction a useful test bed for the stronger idea that Hecke structure might descend to a primitive dynamical Euler product.

The central research question is deliberately narrower than that broad motivation:

> For the frozen level-11 newform time change, do the Hecke cycle-period identities imply either of the scalar, all-s second-variation recurrences $\lambda_p = a_p$ or $\lambda_p = a_p^2$ on the registered finite primitive-owner output multiset?

The answer is no except for four exactly classified kernel groups. The point is not merely that a numerical fit fails. The Hecke correspondence acts on a cycle by producing a finite sum of closed owners of different branch-cycle degrees. Those degrees alter the length kernels in a dynamical-zeta expansion and are not zeta repetitions. At second order, inverse orientations add rather than cancel, but the resulting sum depends on squares of owner periods. A scalar recurrence therefore imposes a family of degree-wise quadratic moments that does not follow from the linear Hecke eigenperiod relation.

This article makes four contributions. First, it fixes the positive time change and proves the conjugacy, orientation, and repetition laws of its period coordinate. Second, it states the cycle-pushforward Hecke theorem with an explicit closed-owner decomposition, separating a Hecke branch-cycle degree from a zeta repetition index. Third, it derives exact first- and second-variation formulas and proves the necessary-and-sufficient quadratic degree-moment criterion. Fourth, it closes the finite classification problem with exact rational homology: all 138 frozen output instances and all 55 source-word/prime groups are classified without a floating-point zero test.

The conclusion is an obstruction, not a positive determinant construction. The exact taxonomy shows that 134 output instances carry genuine nonzero real-period mass, while the only four kernel instances are explained by compact-homology vanishing or real-structure parity. The result neither enumerates every primitive $\Gamma_0(11)$ class nor defines a globally continued dynamical zeta function. It also does not use rational-prime targets or Riemann-zero data. These boundaries are part of the theorem statement rather than qualifications added after computation.

2 Mathematical setting

2.1 The level-11 differential and time change

Let

$$Y_0(11) = \Gamma_0(11)\backslash\mathbb{H}, \qquad f(z) = \eta(z)^2\eta(11z)^2 = \sum_{n\geq 1} a_n q^n, \quad q = e^{2\pi i z}.$$

The form f is the normalized weight-two newform of level 11 with trivial character. The corresponding newspace has dimension one, the coefficient field is $\mathbb{Q}$, and the displayed eta product has $a_1 = 1$. [2, Properties and q-expansion] Define

$$\omega_f = 2\pi i f(z)\,\mathrm{d}z, \qquad \alpha_f = \mathrm{Re}\,\omega_f.$$

The modular transformation rule for a weight-two form implies that ω_f and α_f descend to the quotient. The classical modular-symbol framework identifies such integrals with homological periods on modular curves. [3, pp. 19–25]

Let X_{geo} be the unit-speed geodesic vector field on $T^1Y_0(11)$. Write $a(v) = \alpha_f(v)$ and set

$$\rho_\varepsilon(v) = 1 + \varepsilon a(v), \qquad X_\varepsilon = \frac{X_{\text{geo}}}{\rho_\varepsilon}.$$

Here ρ_ε is the time density, or slowness, and $1/\rho_\varepsilon$ is the speed multiplier. Cusp-form decay makes a bounded, so $|\varepsilon| < \|a\|_\infty^{-1}$ gives a positive time change.

Proposition 2.1 (period variation)**.** *For an oriented closed geodesic γ of base length $\ell(\gamma)$,*

$$T_\varepsilon(\gamma) = \ell(\gamma) + \varepsilon I(\gamma), \qquad I(\gamma) = \int_\gamma \alpha_f.$$

Moreover, $T_\varepsilon(\gamma^r) = rT_\varepsilon(\gamma)$ for every $r \geq 1$.

Proof. Along the original unit-speed orbit, the new time element is $\mathrm{d}t_\varepsilon = \rho_\varepsilon \, \mathrm{d}t$. Integration gives the first formula. An r-fold traversal concatenates r copies of the same path, multiplying both arclength and the one-form integral by r. □

2.2 Owners, orientation, and repetition

A hyperbolic matrix $M \in \Gamma_0(11)$ determines an oriented quotient loop by projecting an oriented segment of its invariant axis from z to Mz. We write $I(M)$ for its α_f period. The next statement fixes the owner used throughout the paper.

Proposition 2.2 (owner laws)**.** *For $C, M \in \Gamma_0(11)$, with M hyperbolic, and every integer $r \geq 1$,*

$$I(CMC^{-1}) = I(M), \qquad I(M^{-1}) = -I(M), \qquad I(M^r) = rI(M).$$

Thus I belongs to an oriented $\Gamma_0(11)$ conjugacy class; inversion and repetition are separate operations.

Proof. The invariance $C^*\alpha_f = \alpha_f$ sends an axis segment for M to one for CMC^{-1} without changing its integral. Reversing a path changes the sign. A path for M^r is the concatenation of r translates of a path for M, and quotient invariance makes the summands equal. □

This distinction is essential. A primitive owner is not replaced by its r-fold traversal in the primitive product, while orientation reversal is an independent primitive orbit for the oriented geodesic flow.

3 Related work and claim position

Manin's modular-symbol construction places weight-two periods in the homology of modular curves, and Merel gives an explicit treatment of Hecke operators on relative homology using the $\mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$ model. [4, Introduction and Sections 1–2] These sources support the arithmetic and homological setting, but they do not assert a primitive dynamical Euler recurrence for the time-changed flow studied here.

Ruelle introduced zeta functions organized by primitive periodic orbits of hyperbolic dynamics, and Fried carefully separated prime periodic orbits from their iterates in comparing Ruelle and Selberg products. [5, Introduction and flow case] [1, Section 2] We use that primitive/repetition bookkeeping, but only as a finite formal expansion. No global convergence or meromorphic-continuation theorem for the present cusped time-changed flow is claimed.

The paper's novelty position is therefore bounded. The Hecke action on homology and standard dynamical-zeta conventions are prior. The local contributions are the owner-level obstruction chain and its exhaustive, source-locked finite homology taxonomy. The article does not claim the first relation between automorphic periods and dynamics, nor does it infer a global result from a finite computation.

4 Hecke cycle ownership

For a prime $p \nmid 11$, choose the left-coset representatives

$$\beta_b = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \ (0 \le b < p), \qquad \beta_\infty = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}.$$

The weight-two normalization is

$$T_p f(z) = pf(pz) + \frac{1}{p} \sum_{b=0}^{p-1} f\left(\frac{z+b}{p}\right),$$

equivalently $T_p \omega_f = \sum_\beta \beta^* \omega_f$. Since the level-11 cusp space is one-dimensional and f is normalized, $T_p f = a_p f$.

Theorem 4.1 (cycle-pushforward eigenperiod)**.** *For every oriented closed cycle C on $Y_0(11)$,*

$$\int_{T_{p,*}C} \omega_f = a_p \int_C \omega_f, \qquad \int_{T_{p,*}C} \alpha_f = a_p \int_C \alpha_f.$$

If C is represented by a hyperbolic $M \in \Gamma_0(11)$, the left side decomposes into a finite sum of explicitly owned closed geodesics.

Proof. Pair the correspondence with the cycle:

$$\int_{T_{p,*}C} \omega_f = \sum_\beta \int_{\beta C} \omega_f = \int_C \sum_\beta \beta^* \omega_f = \int_C T_p \omega_f = a_p \int_C \omega_f.$$

The coefficients a_p are real, so taking real parts proves the second identity.

For the explicit decomposition, right multiplication by M permutes the representative set. There are unique $\pi(j)$ and $\gamma_j \in \Gamma_0(11)$ with

$$\beta_j M = \gamma_j \beta_{\pi(j)}.$$

For a permutation cycle $O = (j, \pi(j), \ldots, \pi^{d-1}(j))$, iteration gives

$$\delta_O := \beta_j M^d \beta_j^{-1} = \gamma_j \gamma_{\pi(j)} \cdots \gamma_{\pi^{d-1}(j)} \in \Gamma_0(11).$$

The projected branch chain closes and is owned by δ_O. Changing the starting branch cyclically conjugates the product in $\Gamma_0(11)$. Summing over cycles reconstructs $T_{p,*}C$. ☐

The theorem is sum-valued. It does not send one primitive orbit to one primitive orbit. A further control makes this limitation sharp: because $X_0(11)$ has genus one, $H^1(X_0(11), \mathbb{R})$ is generated by the real and imaginary parts of ω_f, and T_p acts by the real scalar a_p on both. Hence every smooth closed real one-form extending to $X_0(11)$ satisfies the same cycle-period relation. The identity is structurally correct but is not discriminative evidence for a primitive Euler mechanism.

4.1 Three indices that must remain distinct

The correspondence introduces three superficially similar integers. The first is the branch-cycle degree d_O, the length of an orbit of the permutation π on the $p+1$ coset representatives. The branch starting at j returns to the same sheet only after d_O applications of M, and its closed owner is $\delta_O = \beta_j M^{d_O} \beta_j^{-1}$. Since the fractional-linear action of β_j is an isometry of $\mathbb{H}$, the resulting closed path has length $d_O \ell(M)$. This explains the length multiplication without identifying δ_O with a power of M inside $\Gamma_0(11)$.

The second integer is the primitive-root exponent of δ_O. If $\delta_O = h^q$ for some $h \in \Gamma_0(11)$ and $q > 1$, then the closed path traverses the primitive h-geodesic q times. The frozen root certificates give $q = 1$ for every registered output. Thus $d_O > 1$ can coexist with a primitive output owner: degree records how a Hecke branch closes, whereas a root exponent records repetition of one geodesic in the quotient.

The third integer is the logarithmic-product repetition r in equations (1)–(2). It appears only after a primitive owner has been selected and enumerates its iterates. Replacing d_O by r, or treating the d_O branches as iterates of the source owner, changes both the owner and the coefficient. Finally, two different permutation cycles can in principle yield conjugate output owners. The finite theorem therefore concerns the registered output *multiset*: each correspondence component carries its declared multiplicity even before global cross-instance conjugacy canonicalization. These distinctions explain why a correct linear equality in homology does not automatically become a primitive-orbit product identity.

5 Zeta variations and exact moment obligations

Let $\gamma^\#$ denote an oriented primitive owner, $L = \ell(\gamma^\#)$, and $I = I(\gamma^\#)$. We use two reciprocal finite formal log products:

$$\log Z_R(s,\varepsilon) = \sum_{\gamma^\#}\sum_{r\geq 1} \frac{e^{-sr(L+\varepsilon I)}}{r}, \tag{1}$$

$$\log Z_S^{\mathrm{fr}}(s,\varepsilon) = \sum_{\gamma^\#}\sum_{r\geq 1} \frac{e^{-sr(L+\varepsilon I)}}{r(1-e^{-rL})}. \tag{2}$$

The second convention freezes the transverse multiplier of the base return map. It is a Selberg-type formal product, not the Selberg zeta function of a deformed metric. Every assertion below is exact for a finite owner family and extends termwise only where an infinite family is absolutely and locally uniformly convergent.

Proposition 5.1 (parity of variations)**.** *The first variations of* (1) *and* (2) *are*

$$\partial_\varepsilon \log Z_R(s,0) = -s\sum_{\gamma^\#} I \sum_{r\geq 1} e^{-srL},$$

$$\partial_\varepsilon \log Z_S^{\mathrm{fr}}(s,0) = -s\sum_{\gamma^\#} I \sum_{r\geq 1} \frac{e^{-srL}}{1-e^{-rL}}.$$

If the family is inverse closed, both vanish. For one inverse pair, the second variations are

$$\partial_\varepsilon^2 \log Z_{R,\mathrm{pair}}(s,0) = 2s^2 I^2 \sum_{r\geq 1} re^{-srL},$$

$$\partial_\varepsilon^2 \log Z_{S,\mathrm{pair}}^{\mathrm{fr}}(s,0) = 2s^2 I^2 \sum_{r\geq 1} \frac{re^{-srL}}{1-e^{-rL}}.$$

Proof. Differentiating one repeated term once cancels the repeated-period factor r against the logarithmic coefficient $1/r$. The inverse owner has the same L and period $-I$, so odd terms cancel. The paired summand is

$$\frac{2e^{-srL}\cosh(sr\varepsilon I)}{r}.$$

Its second derivative at zero is $2s^2 r I^2 e^{-srL}$. The frozen stability denominator is independent of ε and is carried through unchanged. □

For a source pair (M,p), let d_O be the Hecke permutation-cycle degree and set

$$Q_d(M,p) = \sum_{O:d_O=d} I(\delta_O)^2.$$

Although δ_O has length $d_O L(M)$, the finite exact root search certifies each registered δ_O as primitive in $\Gamma_0(11)$. Thus d_O is not the zeta repetition r of δ_O.

Theorem 5.2 (quadratic degree-moment criterion)**.** *Let λ_p depend only on p and be fixed independently of M and s. On the finite Hecke output multiset, the inverse-paired identity*

$$\sum_O V^{(2)}(\delta_O; s) = \lambda_p V^{(2)}(M; s)$$

holds for every sufficiently large real s, for either kernel in Proposition 5.1, if and only if

$$Q_1(M,p) = \lambda_p I(M)^2, \qquad Q_d(M,p) = 0 \quad (d > 1).$$

Proof. Put $q = e^{-sL(M)}$. After removing $2s^2$, the Ruelle output has coefficient

$$n \sum_{d|n} \frac{Q_d}{d}$$

at q^n, whereas the source coefficient is $n\lambda_p I(M)^2$. Equality on an interval of q gives equality of all Taylor coefficients. Dividing by n and applying Möbius inversion yields $Q_1 = \lambda_p I(M)^2$ and $Q_d = 0$ for $d > 1$. Conversely these conditions give the identity term by term. In the frozen-stability kernel, every divisor contribution at degree n has the same nonzero factor $(1 - e^{-nL})^{-1}$, so the same inversion applies. □

The linear relation $\sum_O I(\delta_O) = a_p I(M)$ cannot imply this sum-of-squares condition. The obstruction is algebraic before any root-finding experiment is contemplated.

5.1 Why the linear eigenperiod law does not determine the square moments

Assume first that $I(M) \neq 0$ and write

$$u_O = \frac{I(\delta_O)}{I(M)}.$$

The Hecke eigenperiod identity determines the single signed statistic

$$\sum_O u_O = a_p.$$

The second variation instead asks for one nonnegative statistic in every degree bin,

$$\sum_{O:d_O=d} u_O^2 = \begin{cases} \lambda_p, & d = 1, \\ 0, & d > 1. \end{cases}$$

There is no algebraic implication from the first display to the second. Even within one degree bin, replacing a pair (u_1, u_2) by $(u_1 + t, u_2 - t)$ preserves its signed sum while changing the square sum by $2t(u_1 - u_2) + 2t^2$. Across degree bins, the linear relation does not record where any contribution occurs. Because the second moments are sums of rational squares, the condition at a nonunit degree is especially rigid: it holds exactly when every real period in that bin vanishes.

The all-s quantifier is what exposes this rigidity. For a single numerical value of s, distinct degrees can compensate after being weighted by different exponentials. Equality for every sufficiently large s is equality of a convergent power series near $q = 0$ and therefore fixes every coefficient. Möbius inversion then removes the zeta repetitions and leaves the branch-degree moments. The exact test does not fit λ_p to those coefficients: the three laws were frozen first, and the builder merely evaluates the resulting rational identities.

The case $I(M) = 0$ would require a separate unnormalized formulation, since the ratios u_O would be undefined. It does not occur in the registered source panel: Theorem 6.1 and the source ledger certify $k(M) \neq 0$ for all eleven sources. This nondegeneracy is part of the finite claim boundary, not an assumption about all hyperbolic classes of $\Gamma_0(11)$.

Corollary 5.3 (nonunit mass is a scalar-law obstruction). *If $Q_d(M, p) > 0$ for any $d > 1$, then no choice of scalar λ_p can make the all-s second-variation recurrence hold for that source group. This conclusion is independent of the degree-one moment.*

Proof. Theorem 5.2 requires $Q_d = 0$ at every nonunit degree, and λ_p appears only in the degree-one equation. Since Q_d is a sum of real squares, a positive value cannot be canceled by another owner or by changing the scalar. □

This corollary separates structural failure from normalization failure. A group that has the correct degree-one square sum but positive mass at degree two or higher is not a near success awaiting a better Hecke polynomial; its length support is wrong. In contrast, a group with no nonunit mass can still fail because its degree-one square sum is not the predeclared λ_p. The ledger records both flags, which is why the later counts can distinguish the 47 double failures from the four a_p^2 groups whose only obstruction is nonunit support.

6 Exact Schreier homology classifier

The finite taxonomy uses

$$\mathrm{PSL}_2(\mathbb{Z}) = \langle s, r \mid s^2 = r^3 = 1 \rangle$$

and identifies the twelve right cosets of $\Gamma_0(11)$ with $\mathbb{P}^1(\mathbb{F}_{11})$. A deterministic Schreier transversal produces 24 arcs. The rewritten s^2, r^3, and spanning-tree relations give a 35-row rational relation matrix of rank 21. Consequently

$$\dim_{\mathbb{Q}} H_1(Y_0(11), \mathbb{Q}) = 3.$$

In the frozen coordinate basis, the cusp direction is $(-1, 0, 0)$; quotienting its span gives the two-dimensional compact homology of $X_0(11)$.

Complex conjugation on the modular curve is induced by $z \mapsto -\overline{z}$. Exact rewriting on a rational basis gives the involution

$$\tau(x, y, z) = (-x, y + z, -z), \qquad h + \tau h = (0, 2y + z, 0).$$

Set $k(x, y, z) = 2y + z$. The compact $+1$ eigenspace is one-dimensional, and real integration of ω_f factors through it.

Theorem 6.1 (exact period coordinate). *For every frozen source owner M, $k(M) \neq 0$. For every frozen Hecke cycle owner δ,*

$$\frac{\mathrm{Re} \int_\delta \omega_f}{\mathrm{Re} \int_M \omega_f} = \frac{k(\delta)}{k(M)} \in \mathbb{Q}.$$

The real period vanishes exactly when $k(\delta) = 0$. The full complex period vanishes exactly when the compact homology class of δ is zero.

Proof. For a rational homology class h, invariance of the real part under conjugation gives

$$2\,\mathrm{Re} \int_h \omega_f = \int_{h+\tau h} \omega_f.$$

The symmetrized class is $(0, k(h), 0)$ modulo the cusp direction, so all real periods are scalar multiples of the single compact $+1$-eigenperiod. Taking the ratio for $h = [\delta]$ and $[M]$ gives the formula. The builder verifies $k(M) \neq 0$ for all eleven frozen source owners. Finally, a cusp form extends holomorphically to the compactification, so cusp cycles have zero period. On the genus-one compact curve, integration of a nonzero holomorphic differential embeds rational homology in its period lattice; hence zero full complex period is equivalent to zero compact class. □

This theorem replaces numerical near-zero decisions by exact rational algebra. In particular, normalized moments are

$$M_d = \sum_{O:d_O=d} \left(\frac{k(\delta_O)}{k(M)}\right)^2 \in \mathbb{Q}_{\geq 0}.$$

For $d > 1$, $M_d = 0$ if and only if every owner in that bin lies in the exact real-period kernel.

6.1 Kernel semantics and exact-decision hierarchy

The two kernel labels answer different questions. A full complex-period kernel has zero class in compact homology, so both the real and imaginary periods of ω_f vanish. A projection-only kernel has a nonzero compact class but zero $+1$ component under the real involution; its real period vanishes while its complex period does not. Conflating these cases would erase the mechanism behind two of the four exceptional instances. Conversely, a true nonkernel has $k \neq 0$, which is a proof of nonvanishing of the real period, not merely a failure to recognize zero.

The decision hierarchy is intentionally one-way. Matrix identities and congruence membership are checked over the integers. Schreier rewriting and row reduction are performed over $\mathbb{Q}$. Compact-class and k tests are therefore exact equalities. Only after an exact status has been assigned does the verifier compare inherited binary64 period data as a diagnostic. A small numerical residual can never create a kernel row, and a larger residual can never overturn one. Missing coordinates, failed owner reconstruction, or a broken input hash produce a fail-closed error rather than a fourth empirical category.

This hierarchy also clarifies what the enumeration proves. Exhausting 138 locked rows proves a statement about every row of that finite population because the population hash and row count are inputs to the certificate. It does not prove that the population exhausts all primitive conjugacy classes, all positive words, or all Hecke images. The global and finite quantifiers remain explicit throughout.

7 Complete finite taxonomy

The input population was frozen before classification: eleven positive-word source owners, five primes $p \in \{2, 3, 5, 7, 13\}$, 55 word/prime groups, and 138 Hecke permutation-cycle owner instances. The tested laws were $\lambda_p = a_p$, $\lambda_p = a_p^2$, and the predeclared secondary control $\lambda_p = a_p^2 - p$. No group or scalar was selected after viewing the exact taxonomy.

Theorem 7.1 (exhaustive frozen taxonomy)**.** *The 138 owner instances split mutually exclusively into two full complex-period kernels, two nonzero real-projection-only kernels, and 134 true real-projection nonkernels. There are no degenerate or unresolved instances. The per-prime counts are given in Table 1.*

Table 1: Exact classification of the frozen Hecke cycle-owner instances.

p	full kernel	projection-only	true nonkernel	total
2	0	0	18	18
3	0	0	22	22
5	2	2	26	30
7	0	0	30	30
13	0	0	38	38
All	2	2	134	138

Proof. For every locked cycle row, the verifier reconstructs δ_O from $\beta_j M^{d_O} \beta_j^{-1}$, checks exact equality with the stored matrix, determinant one, the lower-left congruence modulo 11, and the finite primitivity certificate. It then converts the matrix into an s, r word, applies exact Schreier rewriting over `fractions.Fraction`, and computes its compact class, conjugate class, and k coordinate. The mutually exclusive decision rule is: compact class zero gives a full kernel; compact class nonzero and $k = 0$ gives a projection-only kernel; $k \neq 0$ gives a true nonkernel; unavailable exact data would fail closed as degenerate. Exhaustive iteration produces the displayed counts and zero degenerate rows. This is a finite proof by exact enumeration; the checked program and data hashes are part of the certificate described in Section 8. □

The four kernels all occur at $p = 5$ and cycle degree five. The words `LRRLRRR` and `LLRLLRLR` have compact-zero output classes. The words `LLLRLLRLR` and `LLLRLRLLR` have nonzero compact classes that are anti-invariant under τ, so their complex periods are nonzero and purely imaginary while their real projections vanish.

Corollary 7.2 (group-level recurrence verdicts)**.** *For each primary law a_p and a_p^2, exactly four of 55 groups satisfy all quadratic degree moments and 51 fail. The four survivors are the $p = 5$ kernel groups above. The secondary control $a_p^2 - p$ fails all 55 groups.*

Proof. For every group and law, the verifier forms the exact rational square sums M_d and subtracts the required moment λ_p at degree one and zero at every nonunit degree. The two primary laws coincide at $p = 5$ because $a_5 = 1$. Exactly the four kernel groups have $M_1 = 1$ and $M_d = 0$ for $d > 1$. For a_p, all 51 failures violate both the degree-one condition and a nonunit condition. For a_p^2, 47 violate both and four satisfy degree one but retain nonunit mass. The control changes the required degree-one scalar and therefore rejects even the four kernel groups. □

The result explains both sides of the earlier numerical audit. The 51 primary failures contain exact nonunit-degree quadratic mass and cannot be repaired by increased precision. The four positives are topological or parity kernels, not evidence of a new prime-to-orbit assignment.

There are two denominators in these statements. The instance denominator 138 counts individual permutation-cycle components and supports the three-way kernel taxonomy. The group denominator 55 counts source-word/prime pairs and is the unit on which a scalar recurrence is either satisfied or rejected. One exceptional instance can affect its whole group, while several instances in a group contribute to the same degree moment. Reporting only an instance success percentage would therefore answer the wrong question. Conversely, reporting only 51 failed groups would hide why the four surviving groups are non-generic. The paired ledgers preserve both levels of ownership.

Nor are the four survivors validation data for a revised law. All occur at the single prime $p = 5$, where the two primary scalars coincide, and all are explained by exact kernel geometry. They contain no evidence that the same recurrence extends to a new source or prime. Under the frozen protocol, they remain classified positives for the tested finite identity while functioning as adversarial controls against an arithmetic interpretation.

8 Certificate and reproducibility method

The Round-8 builder is a Python-standard-library program. Its locked inputs are

- `round4_hecke_cycle_ledger.csv`, with SHA-256 `f906df349b8f1fa2864fed592792e0ff f63ba246a069179b7bd8cfdf46520662`; and
- `round6_quadratic_degree_moment_ledger.csv`, with SHA-256 `f95e1435c9293f8e008cebf80084ea2b 522b76186dbd684b5e3997c5e588edea`.

Exact rational homology and rational square sums decide the theorem; inherited binary64 periods are retained only as cross-checks.

The generated instance ledger has 138 rows, and the group/law ledger has 165 rows. All 165 exact verdicts agree with the earlier numerical statuses, with maximum normalized cross-check residual

$1.9895196601282805 \times 10^{-13}$. This agreement is not used as proof. Eighteen Round-8 unit tests cover input hashes, population counts, the Schreier model, the involution formula on every owner, exact owner rebuilding, primitivity, taxonomy counts, square-sum identities, law verdicts, fail-closed tampering, route boundaries, and deterministic in-memory rebuilding.

Running `experiments/reproduce_round8.sh` performs the tests, generates two isolated four-file trees, compares them byte for byte, and checks the result against the repository artifacts. The two tree hashes are both

`cc36c1f952c9ce89050996f4bb4c9905571f9ef09a0d7115be8a985e02a5621d`.

The computation is therefore a reproducible exact certificate for the registered finite population, not a numerical sample used to infer an unbounded theorem.

The certificate is layered so that each conclusion can be audited at its natural level. The source lock prevents a later change of words, primes, or candidate laws. Owner rebuilding verifies that each stored matrix really arises from the declared source, coset branch, and permutation cycle. The homology layer derives, rather than imports, the compact class and involution coordinate. The moment layer groups exact ratios by branch degree and evaluates all three predeclared laws. Finally, the serializer sorts rows canonically and hashes the complete output tree. Agreement of two isolated trees checks determinism; agreement with the checked-in tree checks repository integrity. Neither check replaces the mathematical proof, but together they bind the proof's finite hypotheses to inspectable artifacts.

Several common failure modes are therefore observable. Changing one input byte breaks the source hash. Dropping or duplicating a row breaks population invariants. Substituting floating-point zero tests changes the schema expected by the validator. Reclassifying a projection-only kernel as a full kernel breaks the compact-class consistency check. Editing a Route verdict without changing the underlying evidence breaks the boundary assertions. The test suite includes representative tampering cases, so reproducibility means more than rerunning a script that always exits successfully.

9 Adversarial controls and Route-A interpretation

The control analysis separates the five Route-A layers. At A0, the level-11 newform supplies genuine arithmetic structure, and no prime table or Riemann-zero table enters the object. Nevertheless, there is no intrinsic rational-prime owner map or natural $\log p$ clock, so the arithmetic relation remains weak. At A1, the period owner and finite Hecke cycle owners are explicit and reproducible, but the positive-word source ledger is not a complete primitive $\Gamma_0(11)$ conjugacy census. The appropriate verdict is therefore weak rather than a certified full primitive-orbit layer.

At A2, no global dynamical determinant was built, no training/validation/test divisor regions were defined, and no argument-principle root count, missing/extra-zero report, cutoff drift, or precision drift was run. The finite log-product obstruction does not substitute for those requirements. A3 fails because no functional equation, gamma factor, analytic continuation, counting law, or Weil-type compression was derived from this candidate. A4 fails because no natural unitary or scattering lift preserving this clock and these weights was constructed. The conservative tuple is

$$(\text{A0\_WEAK\_ARITHMETIC\_RELATION, A1\_WEAK, A2\_FAIL, A3\_FAIL, A4\_FAIL}),$$

with overall status `ROUTE_A_EXPLORATORY`; Route B is not authorized.

Two proves-too-much controls are decisive. First, every compactly extending closed real one-form on the genus-one curve obeys the same scalar Hecke cycle relation, so the linear relation is not selective for α_f. Second, inverse pairing makes every real one-form's first variation vanish and makes its second variation a square. The mechanism is therefore generic within a large control class. The exact $a_p^2 - p$ law further rejects the possibility that the four survivors reflect a flexible Hecke-polynomial choice. These controls support a scoped obstruction theorem, not an RH-like conclusion.

10 Limitations and open obligations

The taxonomy is complete only for the frozen output multiset. It does not deduplicate $\Gamma_0(11)$-conjugate owners across different output instances, enlarge the source word cutoff, or enumerate all primitive conjugacy classes. Although each of the 138 stored instances has an exact finite root certificate, this does not show that there are 138 globally distinct owners. Cross-instance conjugacy canonicalization is the smallest next finite task.

The analytic scope is also local. The formal products are differentiated on finite owner families; no convergence or continuation theorem is proved for the full cusped time-changed flow. A different, genuinely non-scalar Hecke action on a transfer operator is not rejected by Theorem 5.2. The theorem refutes only the registered p-only scalar recurrences on the registered finite Hecke output multiset. Finally, the study provides no rational-prime dictionary, no target-zero comparison, and no quantum operator. These omissions prevent promotion to Route-A A2 and prevent any Route-B inference.

11 Conclusion

The level-11 newform time change has a clean arithmetic period coordinate and an exact Hecke relation, but the correct relation is owned by a cycle pushforward, not by one primitive orbit. Once branch-cycle degree, zeta repetition, orientation, and multiplicity are kept distinct, the tested primitive-Euler interpretation fails. Exact Schreier homology turns this failure into a complete finite theorem: 134 of 138 output instances are genuine nonkernels, and the four remaining instances are fully explained by compact topology or real-structure parity. The resulting paper-level contribution is an exact taxonomy and non-implication theorem. A global determinant or Hilbert–Pólya candidate remains outside the established evidence.

Data and Code Availability

All data used in this study are repository artifacts under `papers/26-level11-newform-time-change/results`. The deterministic source, unit tests, and verify-default reproduction script are under `code` and `experiments`. The manuscript reports the relevant SHA-256 bindings and requires no proprietary or target-side dataset.

Ethics Declaration

This theoretical and computational mathematics study involved no human participants, personal data, animals, clinical material, or field intervention. Institutional ethics approval and informed consent were therefore not applicable.

Author Contributions

Liang Wang: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing–Original Draft, Writing–Review & Editing, Visualization, and Project Administration.

Conflict of Interest

The author declares no conflict of interest.

Funding

This research received no external funding.

AI-Assisted Research Disclosure

AI-assisted tools were used to support research organization, source-metadata checking, code development, mathematical exposition, and manuscript drafting. The author retains responsibility for the definitions, theorem statements, proofs, computations, citations, claim boundaries, and final text. Exact computational claims are accompanied by deterministic tests, source locks, and reproducibility receipts; AI output is not treated as independent evidence.

References

[1] David Fried. The zeta functions of Ruelle and Selberg. I. *Annales scientifiques de l'École Normale Supérieure*, 19(4):491–517, 1986. doi: 10.24033/asens.1515. URL https://www.numdam.org/item/ASENS_1986_4_19_4_491_0/.

[2] LMFDB Collaboration. Newform orbit 11.2.a.a. LMFDB, 2026. URL https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/11/2/a/a/. Accessed 28 August 2026.

[3] Yuri I. Manin. Parabolic points and zeta-functions of modular curves. *Mathematics of the USSR-Izvestiya*, 6(1):19–64, 1972. doi: 10.1070/IM1972v006n01ABEH001867. URL https://www.mathnet.ru/eng/im2290.

[4] Loïc Merel. Opérateurs de Hecke pour $\gamma_0(n)$ et fractions continues. *Annales de l'Institut Fourier*, 41 (3):519–537, 1991. doi: 10.5802/aif.1264. URL https://numdam.org/articles/10.5802/aif.1264/.

[5] David Ruelle. Zeta-functions for expanding maps and Anosov flows. *Inventiones Mathematicae*, 34 (3):231–242, 1976. doi: 10.1007/BF01403069. URL https://link.springer.com/article/10.1007/BF01403069.